

## PETISTRASK, Sweden

WMO: 022820

Lat: 64.5659N

Lon: 19.6943E

Elev: 258

StdP: 98.26

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -29.8      | -26.3 | -33.0                         | 0.2 | -29.7 | -29.0 | 0.3 | -26.2 | 5.5                   | -3.0 | 5.1 | -3.4 | 0.2                   | 270  | 0.620 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.8                    | 25.3            | 16.9 | 23.3 | 16.2 | 21.4 | 15.1 | 18.4                | 23.2 | 17.2 | 21.6 | 16.1 | 20.2 | 1.7                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.5                            | 12.1 | 20.2 | 15.4 | 11.2 | 19.0 | 14.3 | 10.5 | 17.9 | 52.9          | 23.3 | 49.2 | 21.7 | 46.0 | 20.4 | 22.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 5.4               | 4.5  | 4.0 | -33.0                      | 27.5 | 3.9                | 2.4 | -35.9                                              | 29.3 | -38.2      | 30.7 | -40.4      | 32.0 | -43.2      | 33.8 |
|                   |      |     | -33.4                      | 19.8 | 3.8                | 1.5 | -36.2                                              | 20.9 | -38.4      | 21.7 | -40.5      | 22.6 | -43.3      | 23.6 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 1.7   | -10.7 | -10.4 | -5.6  | 0.5   | 7.2   | 12.0  | 15.2  | 13.4  | 8.2   | 1.6   | -3.7  | -7.9  | (6)  |
| (7)  |                                                                                  | DBStd       | 10.29 | 6.97  | 7.47  | 5.68  | 3.82  | 3.93  | 3.36  | 3.01  | 3.04  | 3.22  | 4.12  | 5.28  | 6.89  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3407  | 643   | 572   | 483   | 286   | 104   | 17    | 1     | 6     | 71    | 261   | 412   | 553   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 6076  | 901   | 805   | 742   | 536   | 344   | 192   | 107   | 154   | 303   | 518   | 662   | 812   | (9)  |
| (10) |                                                                                  | CDD10.0     | 387   | 0     | 0     | 0     | 0     | 18    | 76    | 162   | 112   | 18    | 0     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 13    | 0     | 0     | 0     | 0     | 0     | 2     | 9     | 3     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 170   | 0     | 0     | 0     | 0     | 4     | 36    | 102   | 28    | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 22    | 0     | 0     | 0     | 0     | 0     | 4     | 17    | 1     | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 1.6   | 1.4   | 1.5   | 1.9   | 1.9   | 2.0   | 1.8   | 1.5   | 1.4   | 1.4   | 1.7   | 1.5   | 1.5   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 598   | 35    | 33    | 28    | 30    | 39    | 62    | 99    | 75    | 57    | 51    | 48    | 41    | (15) |
| (16) |                                                                                  | PrecMax     | 744   | 70    | 72    | 56    | 57    | 61    | 96    | 152   | 126   | 111   | 93    | 84    | 78    | (16) |
| (17) |                                                                                  | PrecMin     | 484   | 4     | 6     | 9     | 7     | 9     | 25    | 20    | 22    | 20    | 16    | 21    | 17    | (17) |
| (18) |                                                                                  | PrecStd     | 88    | 14    | 16    | 13    | 15    | 17    | 20    | 36    | 33    | 32    | 22    | 21    | 15    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 3.6   | 5.2   | 8.0   | 16.0  | 23.7  | 27.0  | 28.8  | 26.1  | 19.2  | 12.2  | 8.0   | 4.1   | (19) |
| (20) |                                                                                  | MCWB        | 2.4   | 3.1   | 4.0   | 8.6   | 13.8  | 17.1  | 19.2  | 18.4  | 14.3  | 9.6   | 6.9   | 2.8   | 2.8   | (20) |
| (21) |                                                                                  | 2%          | DB    | 1.4   | 3.6   | 5.6   | 12.0  | 21.0  | 23.8  | 26.1  | 23.8  | 17.1  | 10.2  | 5.9   | 2.9   | (21) |
| (22) |                                                                                  | MCWB        | 0.7   | 1.9   | 2.7   | 6.3   | 12.7  | 15.3  | 17.9  | 17.1  | 13.0  | 8.6   | 5.2   | 2.1   | 2.1   | (22) |
| (23) |                                                                                  | 5%          | DB    | 0.2   | 1.7   | 4.2   | 9.5   | 18.0  | 21.2  | 23.9  | 21.8  | 15.2  | 8.8   | 4.0   | 1.8   | (23) |
| (24) |                                                                                  | MCWB        | -0.2  | 0.5   | 1.7   | 4.8   | 11.3  | 14.0  | 16.7  | 16.2  | 12.0  | 7.5   | 3.4   | 1.2   | 1.2   | (24) |
| (25) |                                                                                  | 10%         | DB    | -1.0  | 0.0   | 2.6   | 7.5   | 15.2  | 19.1  | 22.0  | 19.5  | 13.7  | 7.6   | 2.8   | 0.7   | (25) |
| (26) |                                                                                  | MCWB        | -1.4  | -0.7  | 0.7   | 3.5   | 9.3   | 12.9  | 15.9  | 15.2  | 11.3  | 6.5   | 2.3   | 0.3   | 0.3   | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 2.5   | 3.3   | 4.2   | 8.8   | 14.8  | 17.9  | 20.5  | 19.5  | 15.2  | 10.2  | 7.1   | 3.1   | (27) |
| (28) |                                                                                  | MDCB        | 3.3   | 4.9   | 7.4   | 14.9  | 21.2  | 25.2  | 26.2  | 24.3  | 18.1  | 11.1  | 8.0   | 3.7   | 3.7   | (28) |
| (29) |                                                                                  | 2%          | WB    | 0.8   | 2.0   | 3.0   | 6.9   | 13.3  | 16.1  | 18.8  | 18.0  | 13.7  | 8.8   | 5.2   | 2.1   | (29) |
| (30) |                                                                                  | MDCB        | 1.2   | 3.5   | 5.3   | 11.5  | 20.0  | 22.0  | 23.8  | 22.1  | 15.9  | 9.8   | 5.8   | 2.7   | 2.7   | (30) |
| (31) |                                                                                  | 5%          | WB    | -0.2  | 0.7   | 1.9   | 5.2   | 11.6  | 14.8  | 17.7  | 16.8  | 12.7  | 7.7   | 3.5   | 1.2   | (31) |
| (32) |                                                                                  | MDCB        | 0.1   | 1.5   | 3.6   | 8.8   | 17.3  | 19.7  | 22.5  | 20.4  | 14.5  | 8.6   | 3.9   | 1.6   | 1.6   | (32) |
| (33) |                                                                                  | 10%         | WB    | -1.4  | -0.7  | 0.9   | 3.8   | 9.8   | 13.6  | 16.7  | 15.7  | 11.7  | 6.6   | 2.3   | 0.4   | (33) |
| (34) |                                                                                  | MDCB        | -1.0  | 0.0   | 2.3   | 7.0   | 14.3  | 18.0  | 20.9  | 18.9  | 13.1  | 7.5   | 2.7   | 0.7   | 0.7   | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 8.8   | 10.1  | 10.9  | 10.2  | 10.2  | 9.7   | 9.8   | 9.2   | 8.4   | 6.0   | 5.5   | 7.9   | 7.9   | (35) |
| (36) |                                                                                  | 5% DB       | MCDBR | 7.9   | 8.6   | 10.8  | 13.8  | 15.7  | 13.6  | 13.3  | 13.2  | 10.8  | 7.3   | 4.9   | 6.5   | (36) |
| (37) |                                                                                  | MCWBR       | 7.6   | 7.6   | 8.4   | 8.5   | 8.2   | 6.4   | 6.1   | 6.8   | 7.0   | 5.6   | 4.4   | 6.3   | 6.3   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 7.9   | 8.1   | 9.6   | 12.7  | 14.8  | 12.2  | 12.1  | 11.0  | 9.6   | 6.5   | 4.7   | 6.4   | (38) |
| (39) |                                                                                  | MCWBR       | 7.6   | 7.3   | 7.5   | 8.1   | 8.1   | 6.2   | 6.0   | 6.1   | 6.6   | 5.4   | 4.3   | 6.1   | 6.1   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.194 | 0.266 | 0.296 | 0.329 | 0.335 | 0.337 | 0.349 | 0.337 | 0.307 | 0.290 | 0.337 | 0.351 | 0.351 | (40) |
| (41) |                                                                                  | taud        | 1.972 | 2.187 | 2.252 | 2.346 | 2.473 | 2.494 | 2.473 | 2.501 | 2.568 | 2.434 | 2.695 | 2.728 | 2.728 | (41) |
| (42) |                                                                                  | Ebn at Noon | 359   | 641   | 788   | 846   | 877   | 881   | 856   | 835   | 794   | 629   | 340   | 174   | 174   | (42) |
| (43) |                                                                                  | Edh at Noon | 24    | 56    | 82    | 96    | 93    | 93    | 93    | 82    | 62    | 45    | 21    | 14    | 14    | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.14  | 0.75  | 2.27  | 3.95  | 4.79  | 5.29  | 4.88  | 3.71  | 2.20  | 0.88  | 0.21  | 0.05  | 0.05  | (44) |
| (45) |                                                                                  | RadStd      | 0.01  | 0.12  | 0.21  | 0.30  | 0.57  | 0.44  | 0.44  | 0.54  | 0.36  | 0.28  | 0.15  | 0.04  | 0.00  | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (44 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page